

Corrected Proof

Optimal Error Exponent for Complementary Paley Graphs

Source: Proactively Detecting Structure Information

via Boolean Multiaccess Channels

--- A Correction to Section V ---

Theorem (Corrected Version, Section V)

Let $N = w^2$ (w an odd prime), and G_1, G_2 be complementary

equally-weighted Paley graphs. Let $\{D\}X^*$ be a maximum independent

set of G_1 with $|X^*| = \sqrt{N} = w$.

Then:

(1) The optimal codebook is deterministic:

$$Q(X^*) = 1, Q(X) = 0 \text{ for } X \neq X^*.$$

(2) If G_1 is activated, then $p_1 = 1$ and the output y is

deterministic (always 1).

(3) If G_2 is activated, then $p_2 = 2/(w+1)$.

(4) The optimal error exponent is:

$$D^* = \Theta(1/\sqrt{N}),$$

or equivalently:

$$D^* = \log((2(\sqrt{N}+1))/(\sqrt{N}+3)).$$

Proof Step 1: Optimal Codebook is Deterministic for $K = 2$

From the optimization problem (Eqs. 9-10), the constraints provide

K equations plus the probability normalization constraint

$\sum_x Q(x) = 1$. Hence the degrees of freedom are $K-1$.

For $K = 2$, there is only 1 degree of freedom. A single codeword

suffices to achieve optimality. Therefore the optimal solution must

be a deterministic codebook:

$$Q(x^*) = 1.$$

QED.

Proof Step 2: Determining the Optimal Codeword X^*

Let X^* be a maximum independent set of G_1 , with $|X^*| = w$.

Claim: $\{D\}X^*$ is the optimal codeword.

Reason:

Let X be any independent set of G_1 . Then $p_1(X) = 1$

(since X has no internal edges, $\alpha(X, G_1) = 0$).

The probability under G_2 is $p_2(X) = 1 - \alpha(X, G_2)$.

To maximize separation, we need to minimize $p_2(X)$, i.e.,

maximize $\alpha(X, G_2)$.

Since G_1 and G_2 are complementary, X is an independent set

of G_1 iff X induces a complete subgraph in G_2 (every pair

of nodes in X is adjacent in G_2).

Hence the larger $|X|$, the higher $\alpha(X, G_2)$.

Therefore the maximum independent set X^* maximizes $\alpha(X^*, G_2)$,

minimizes p_2 , and maximizes Chernoff information.

QED.

Proof Step 3: Exact Calculation of $\alpha(X^*, G_2)$

Lemma: $|E(G_2[X^*])| = w(w-1)/2$.

Proof: Since $\{D\}X^*$ is an independent set in $\{D\}G_1$, no edge between

any two nodes of X^* belongs to G_1 . Because G_1 and G_2 are

complementary, every such edge $\{i, j\}$ belongs to G_2 . Thus X^*

induces a complete graph K_w in G_2 , with $w(w-1)/2$ edges.

QED.

Calculation of $\alpha(\{D\}X^*, \{D\}G_2)$:

The total number of edges in a Paley graph is

$$|E(G_k)| = N(N-1)/4 = w^2(w^2-1)/4.$$

$$\alpha(X^*, G_2) = 2|E(G_2[X^*])| / |E(G_2)|$$

$$= 2 * [w(w-1)/2] / [w^2(w^2-1)/4]$$

$$= 4 / [w(w+1)].$$

By Eq. (7): $p_2 = 1 - \alpha(X^*, G_2) = 2/(w+1)$.

QED.

Proof Step 4: Error Exponent

Two hypotheses:

$$H_1: Y \text{ Bern}(1), p_1 = 1$$

$$H_2: Y \text{ Bern}(p_2), p_2 = 2/(w+1)$$

The Chernoff information for distinguishing $\text{Bern}(1)$ from $\text{Bern}(q)$:

$$C(1, q) = -\log((1+q)/2).$$

Substituting $q = p_2 = 2/(w+1)$:

$$D^* = -\log((1 + 2/(w+1)) / 2)$$

$$= -\log((w + 3)/(2(w+1)))$$

$$= \log(2(w + 1)/(w+3)).$$

Asymptotic expansion (for large w):

$$2(w + 1)/(w+3) = 2 - 4/w + O(1/w^2)$$

$$D^* = \log(2) - 2/w + O(1/w^2)$$

$$D^* = \Theta(1/w).$$

Since $w = \sqrt{N}$:

$$D^* = \Theta(1/\sqrt{N})$$

QED.

Conclusion

Original claim (Section V Theorem):

"The optimal exponent is shown to be $O(1/N)$."

Corrected claim:

"The optimal exponent is shown to be $\Theta(1/\sqrt{N})$,

or equivalently: $\log((2(\sqrt{N}+1))/(\sqrt{N}+3))$."

This correction does NOT affect the paper's core contributions:

[*] Model formulation and optimization framework

[*] Detection feasibility via error exponent analysis

[*] $K = 2$ optimal codebook is deterministic ($Q(X^*) = 1$)

[*] Maximum independent set is the optimal codeword

Only the quantitative exponent order requires revision:

$O(1/N) \rightarrow \Theta(1/\sqrt{N})$